

1. (3 pts) Choose the best option to describe each matrix below. Your choices are:

- Reduced echelon form
- Echelon form but not reduced echelon form
- Not echelon form.

a) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 3 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

2. (2 pts) Either explain carefully in words or show what you would do for the first two steps in row reducing the matrix A without using a calculator.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 0 & 1 & 3 \end{bmatrix}$$

3. (3 pts) Is the vector $\begin{bmatrix} 7 \\ -9 \\ 12 \end{bmatrix}$ in the span of the vectors $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$

(4 pts) Given that $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 3 & 0 \\ 1 & 4 & 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}$ write the matrix equation $A\mathbf{x} = \mathbf{b}$

a) as a system of equations

b) as a vector equation.

5. (3 pts) Describe the **type** of solution set for the system $A\mathbf{x} = \mathbf{b}$ given the **augmented** matrix

$[A \ \mathbf{b}]$

a) $\begin{bmatrix} 2 & 3 & 1 & 2 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

b) $\begin{bmatrix} 2 & 3 & 1 & 2 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 2 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 0 \end{bmatrix}$